Speech and Audio Processing

07 - self-supervised learning

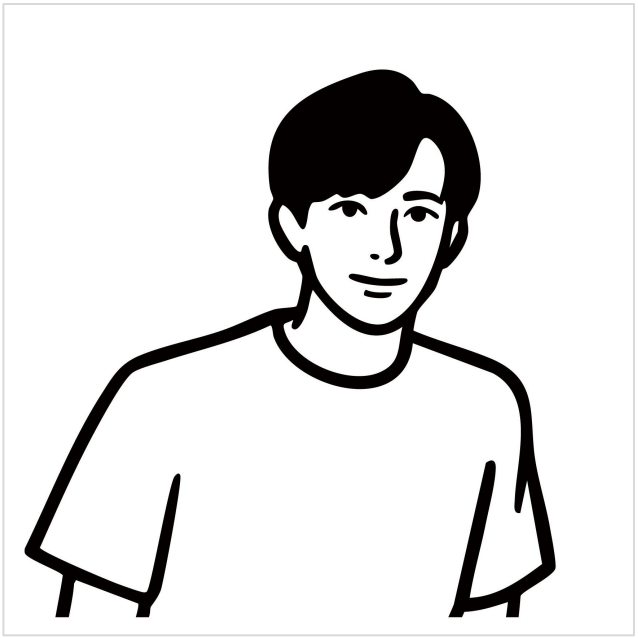
Shinnosuke Takamichi / 高道 慎之介

Takamichi Laboratory
慶應義塾大学 高道研究室

Class schedule (tentative)

01	Oct. 02	Introduction
02	Oct. 09	Speech production / 音声生成
03	Oct. 16	Speech analysis and deep learning / 音声分析と深層学習
04	Oct. 23	Speech synthesis / 音声合成
05	Oct. 30	Speech recognition / 音声認識
06	Nov. 6	**Guest Talk (acoustic signal processing) / ゲストトーク (音響信号処理)**
07	**Nov. 13**	**Self-supervised learning / 自己教師あり学習**
08	Nov. 27	Foundation model / 基盤モデル
09	Dec. 04	**Guest Talk (music information processing) / ゲストトーク (音楽情報処理)**
10	Dec. 11	Speech system building / 音システム構築
11	Dec. 18	Speech system building / 音システム構築
12	Dec. 25	Speech system building / 音システム構築
13	Jan. 08	Speech system building / 音システム構築
14	Jan. 22	Final presentation / 最終発表

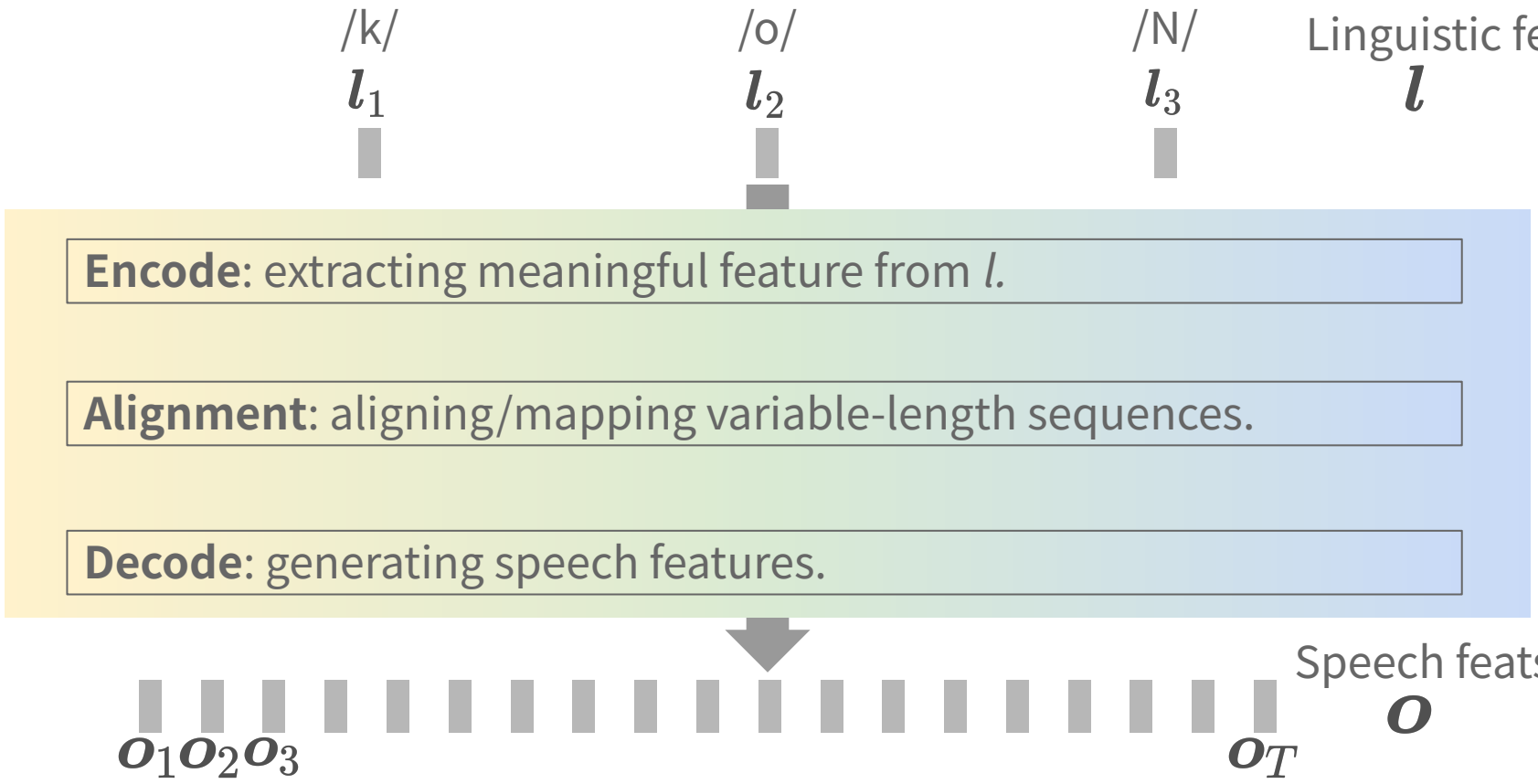
Last week: guest talk



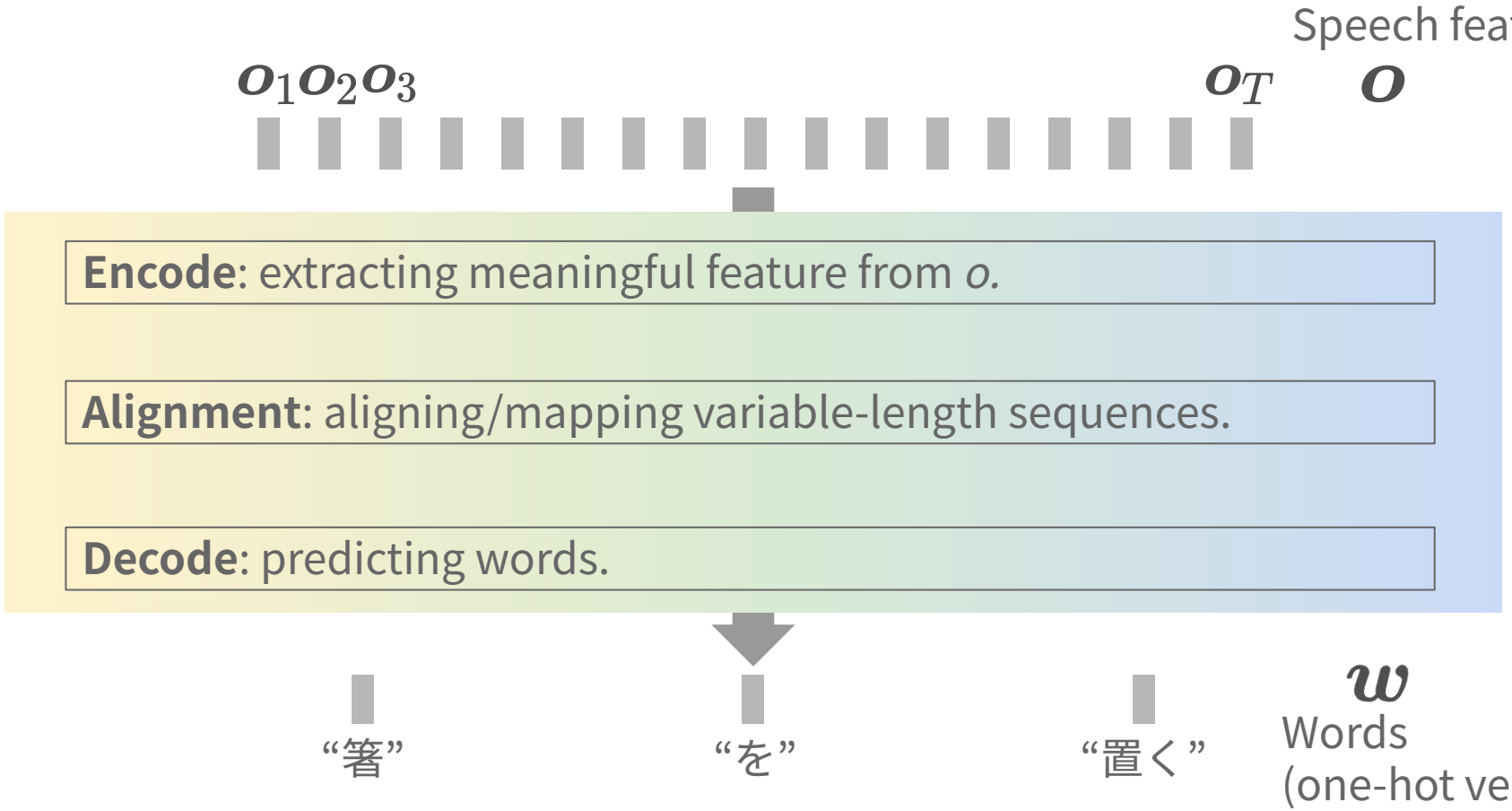
- Taishi Nakashima / 中嶋 大志, Ph.D
  - Earned his degree in 2024.
- Assistant Professor, Tokyo Metropolitan University / 東京都立大学 助教
- Acoustic signal processing / 音響信号処理
- https://taishi.org/

Review of last week’s content / 前回の復習

Inner steps of text-to-speech

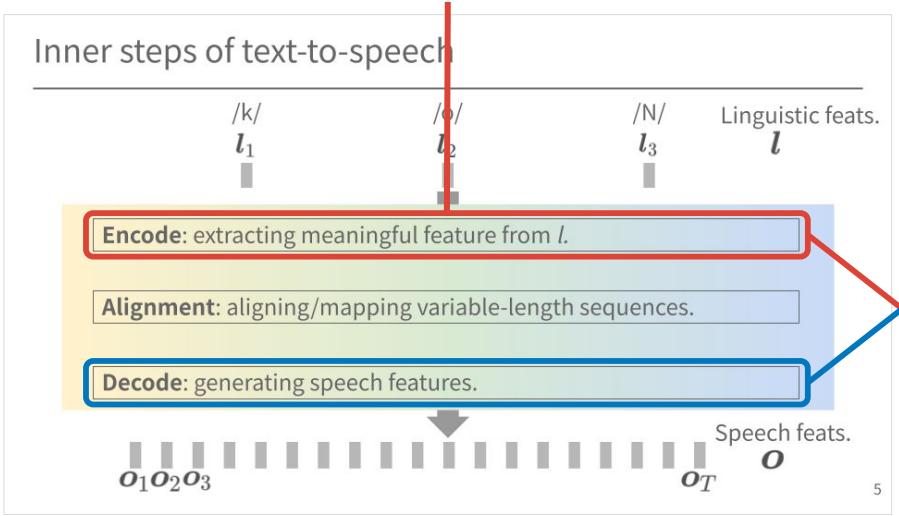


Inner steps of speech-to-text

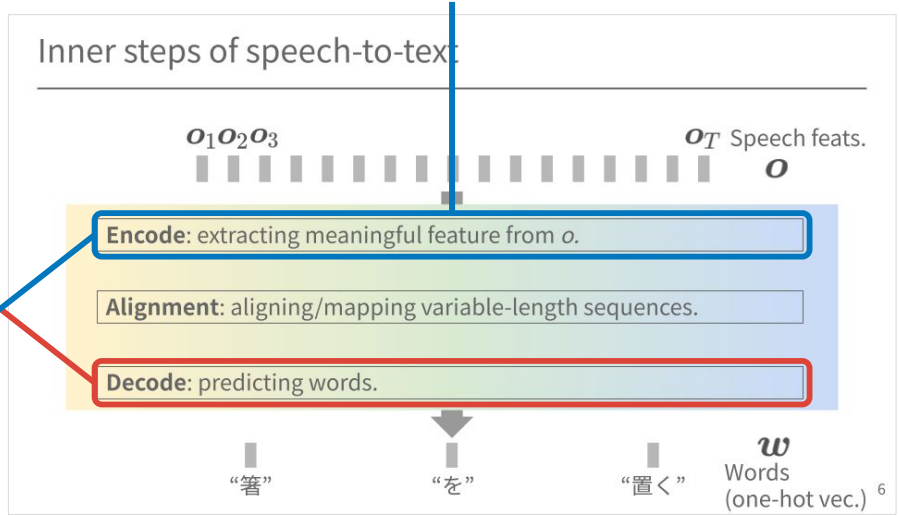


What encoders and decoders learn

Learn to understand/generate **text**.



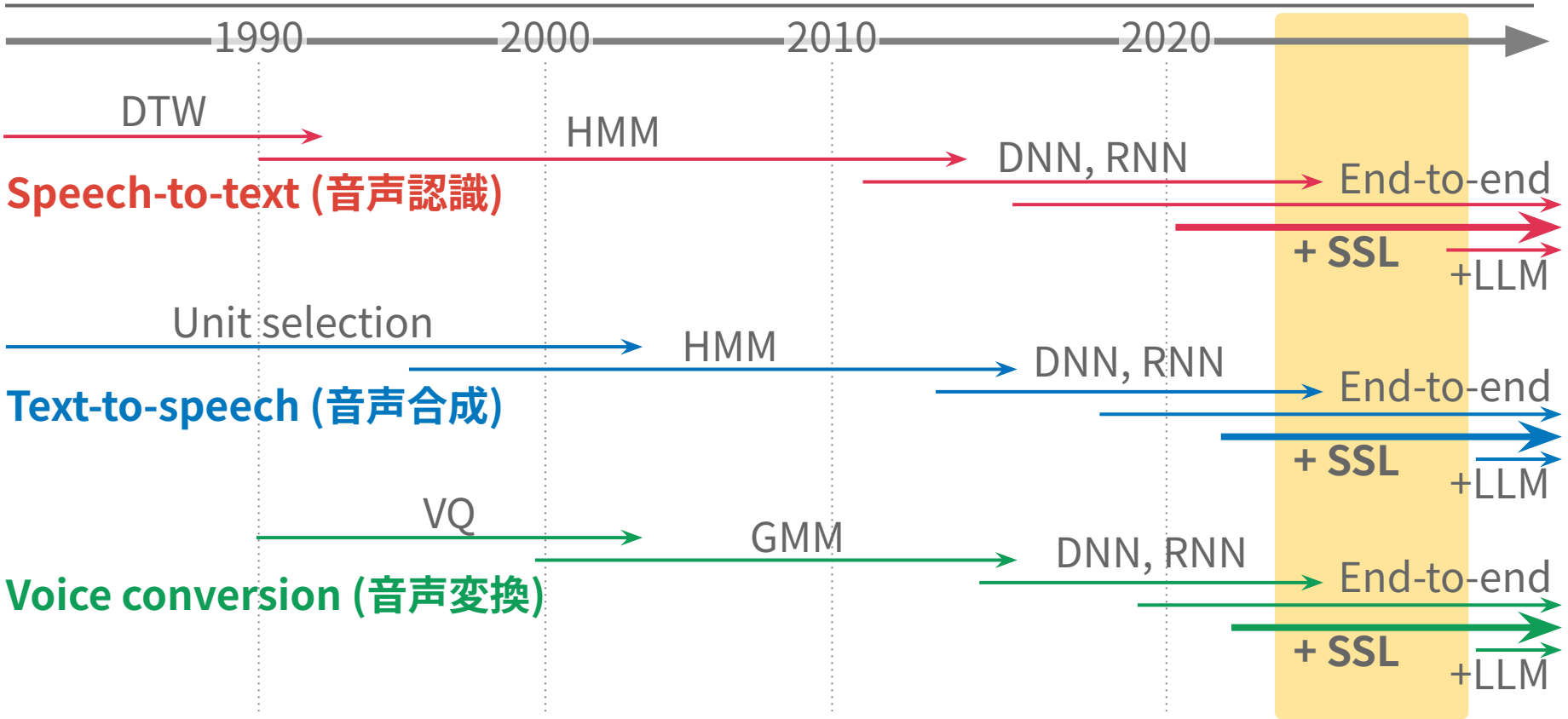
Learn to understand/generate **speech**.



Models are trained on paired text-speech data. -> small data size

→ Can we **pre-train** models on speech-only (or text-only data)?

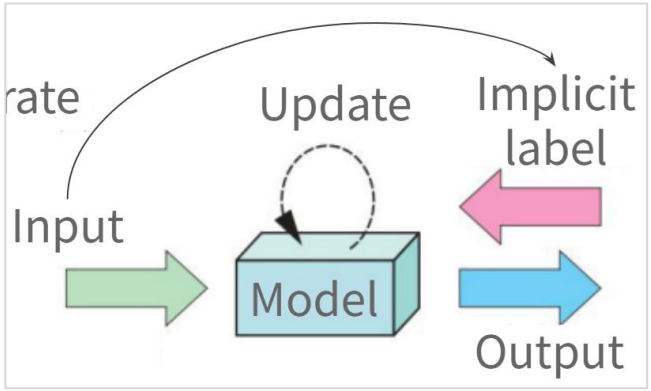
History of speech-based systems

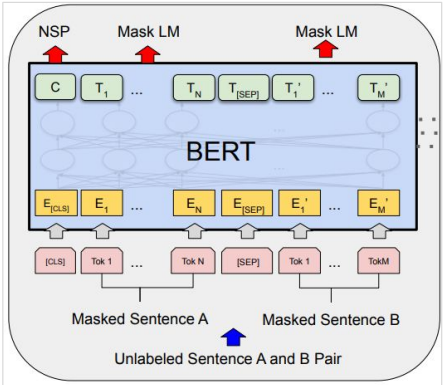


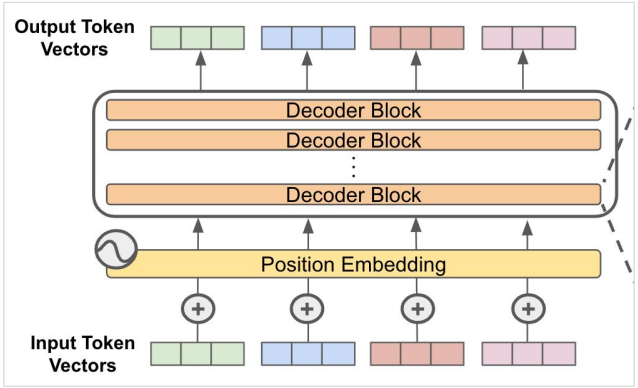
* DTW: dynamic time warping, HMM: hidden Markov model, DNN: deep neural network, RNN: recurrent neural network
VQ: vector quantization, GMM: Gaussian mixture model, SSL: self-supervised learning, LLM: large language model

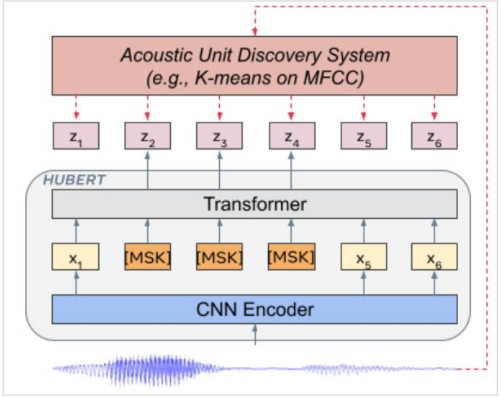
Today's content

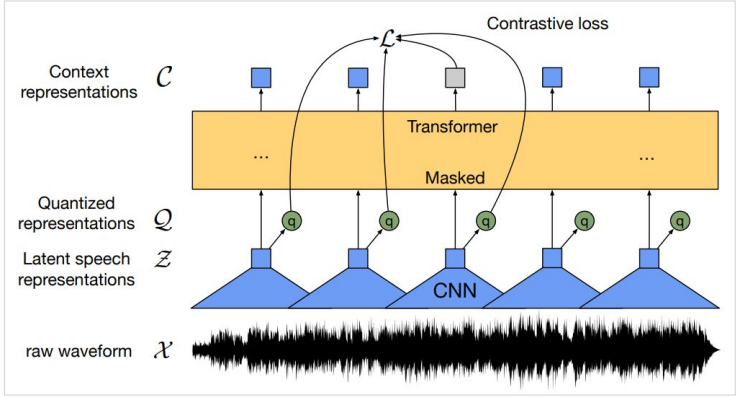
Pretraining models on speech-only data.







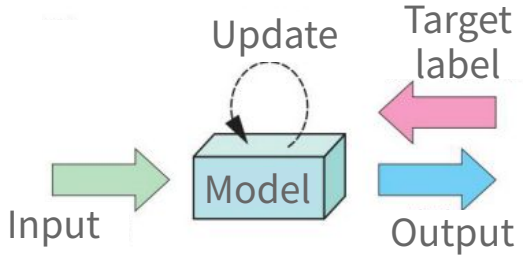




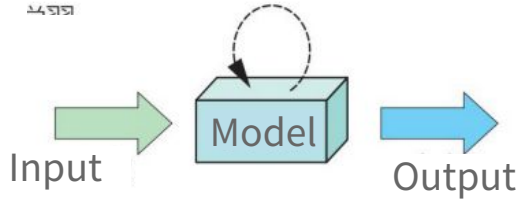
What is self-supervised learning (SSL)?

Several types of machine learning

- **Supervised learning / 教師あり学習**
  - Using labeled data
  - Regression, classification, etc.



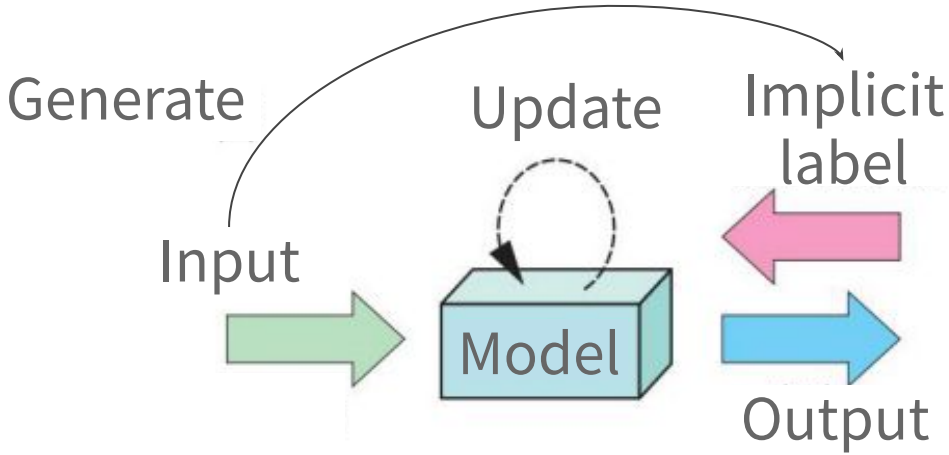
- **Unsupervised learning on unlabeled data / 教師なし学習**
  - Using unlabeled data
  - Clustering, representation learning, etc.



- Weakly-supervised learning / 弱教師あり学習
- Semi-supervised learning / 半教師あり学習
- Reinforcement learning / 強化学習
- Meta learning / メタ学習

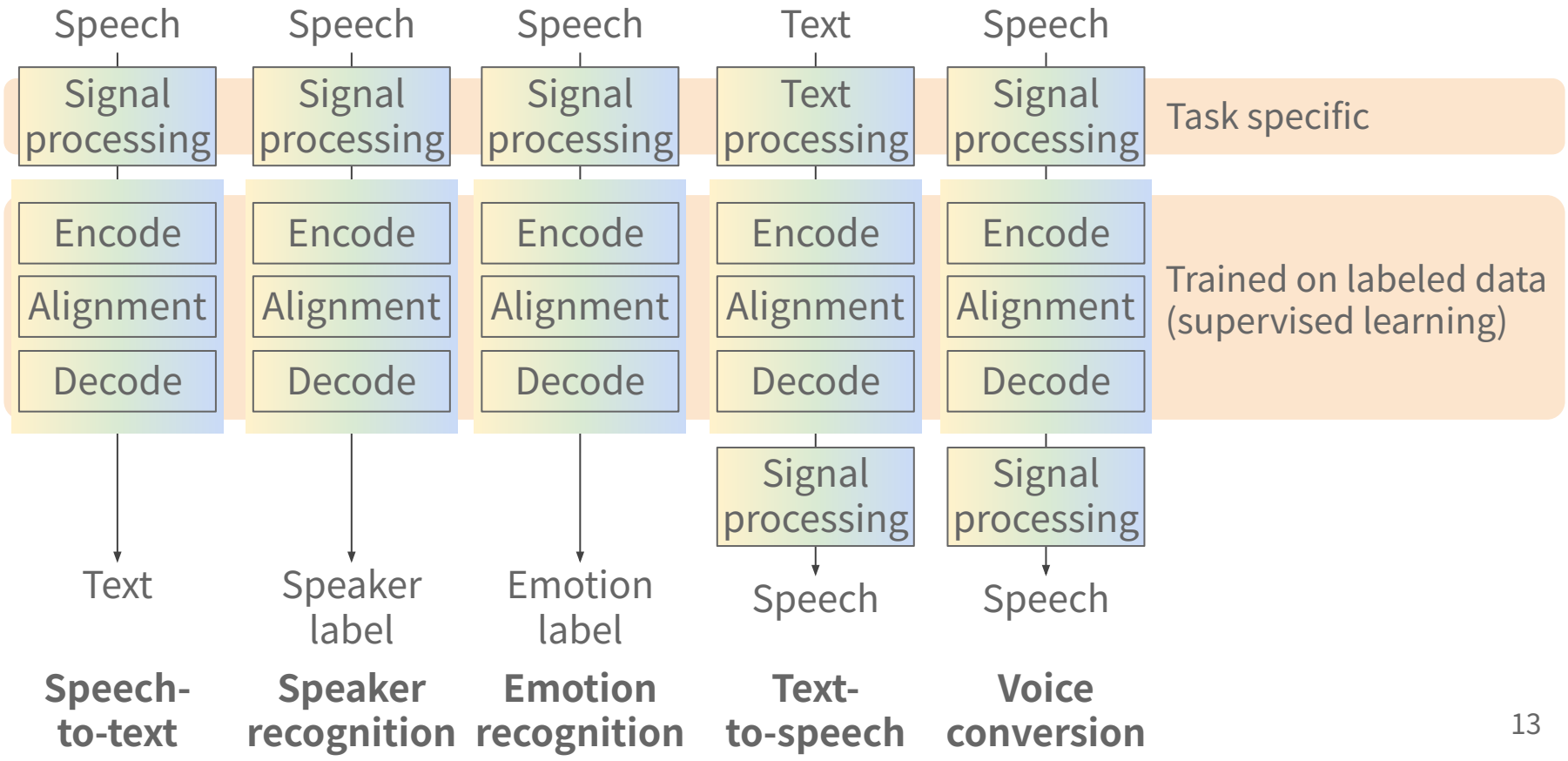
Self-supervised learning (SSL) / 自己教師あり学習

“Rather than relying on labeled datasets for supervisory signals, self-supervised models generate implicit labels from unstructured data.”

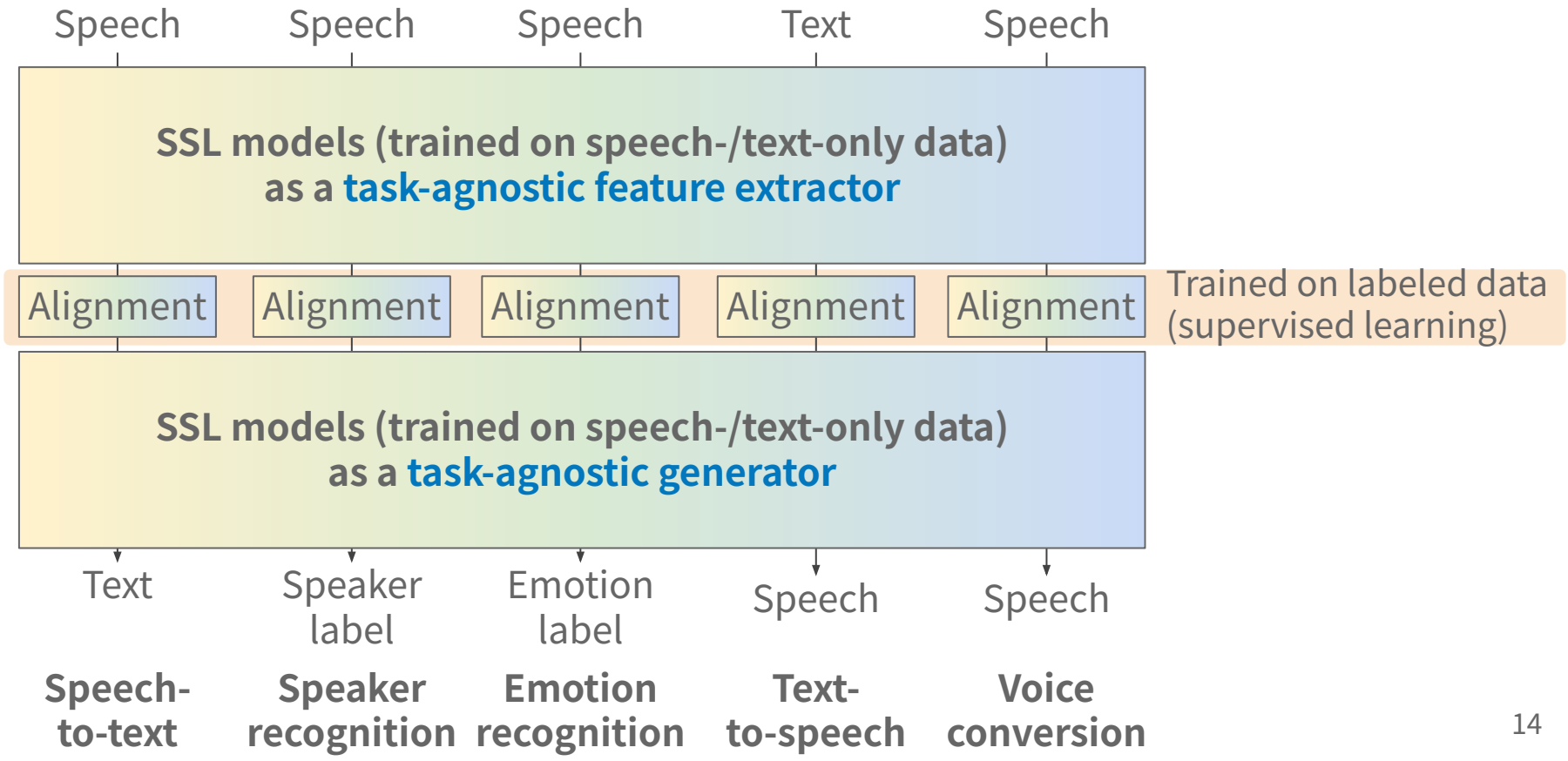


https://www.ibm.com/think/topics/self-supervised-learning

Before SSL era



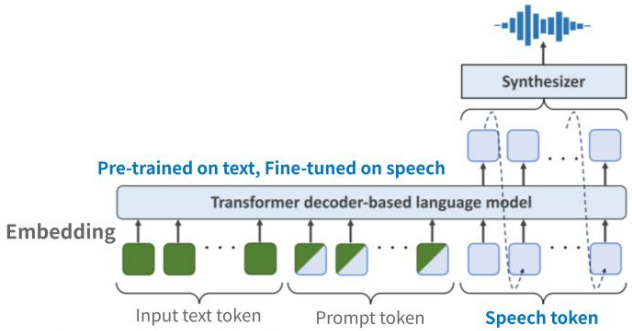
SSL era


Speech
Speech
Speech
Text
Speech
SSL models (trained on speech-/text-only data)
as a task-agnostic feature extractor
Alignment
Alignment
Alignment
Alignment
Alignment
Trained on labeled data
(supervised learning)
SSL models (trained on speech-/text-only data)
as a task-agnostic generator
Text
Speaker label
Emotion label
Speech
Speech
Speech-to-text
Speaker recognition
Emotion recognition
Text-to-speech
Voice conversion

LLM is one kind of SSL models

Typical LLM-based TTS

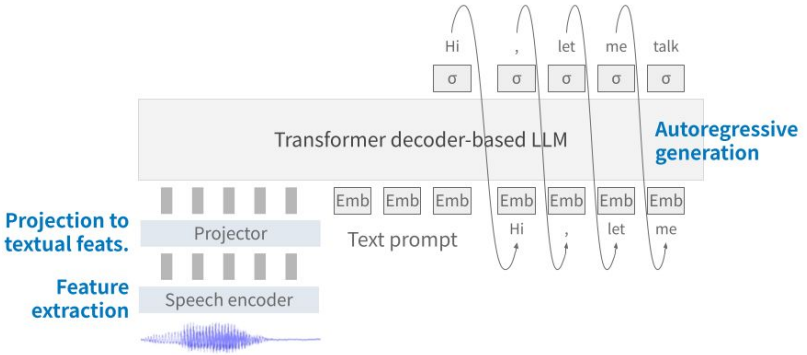
Predicts next speech token using a LLM.



https://www.jstage.jst.go.jp/article/jasj/81/10/81_719/_pdf/-char/ja

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Typical architecture of LLM-based ASR



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Basic architectures and training

Three types of self-supervised learning (for speech)

- **Contrastive approaches / 対照型**
  - Create positive and negative samples.
  - Move the positive sample close and the negative sample far.

- **Predictive approaches / 予測型**
  - (Typically) drop features.
  - Predict the dropped features from undropped ones.

- **Generative approaches / 生成型**
  - Adopt generative models.
  - (Re)generate features via the generative models.

wav2vec 2.0 (2020)

Overview

- Randomly mask CNN outputs.
- Quantize the CNN outputs.
- Move Transformer outputs closer to the CNN outputs.

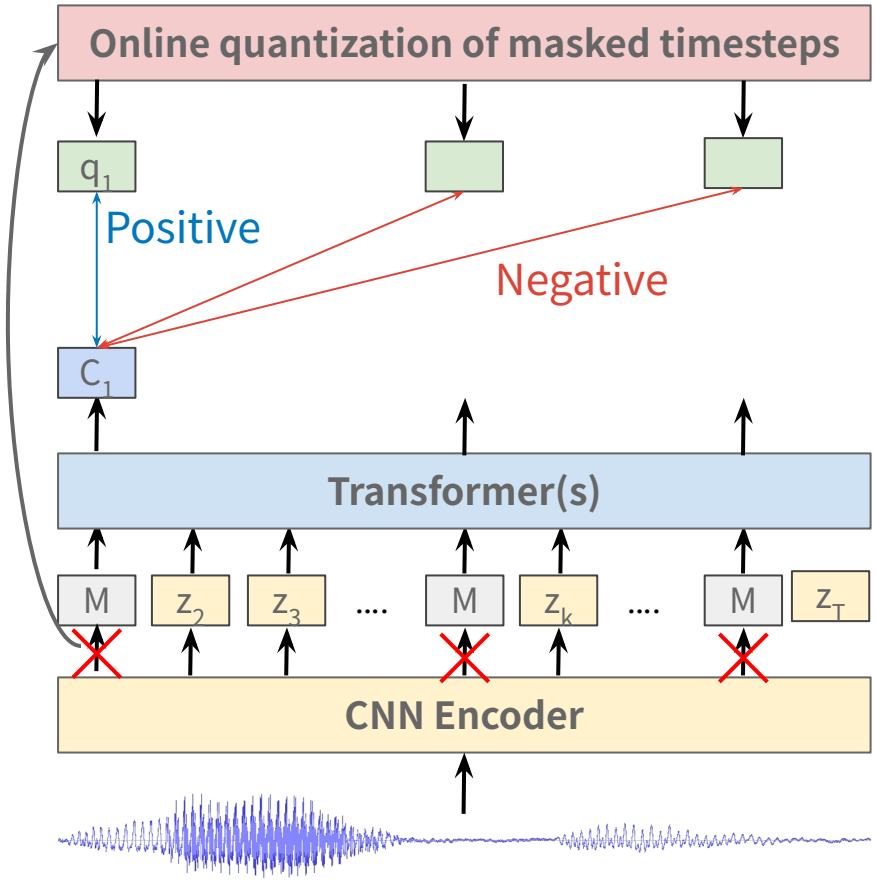
  → Maximize similarity betw. pos.

Training objective

Vector similarity to pos
(i.e., move the pos-pair closer)

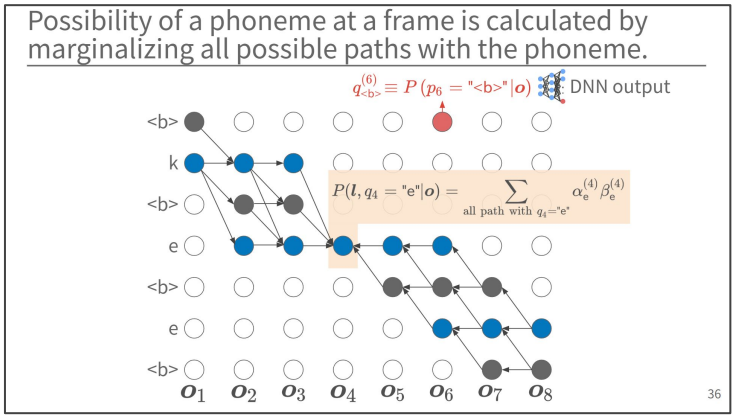
$$\mathcal{L}_m = -\log \frac{\exp(sim(\mathbf{c}_t, \mathbf{q}_t)/\kappa)}{\sum_{\tilde{\mathbf{q}} \sim \mathbf{Q}_t} \exp(sim(\mathbf{c}_t, \tilde{\mathbf{q}})/\kappa)}$$

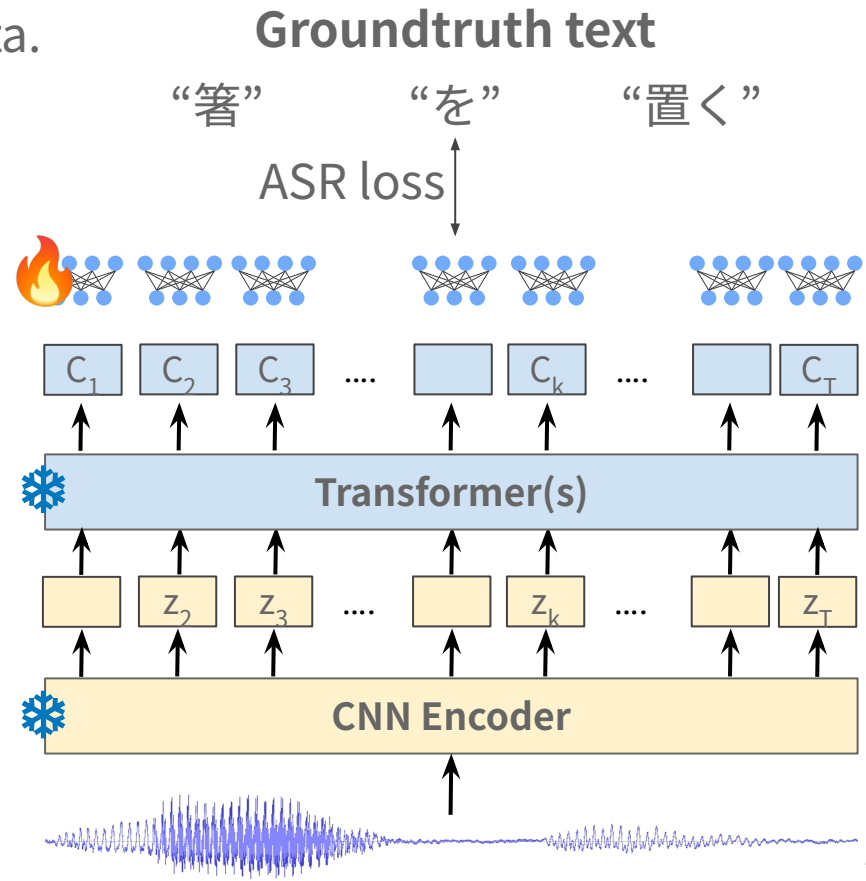
Vector similarities to pos and negs



wav2vec 2.0 → ASR finetuning

- Pretrains wav2vec on speech-only data.
- Adds a linear layer on top of wav2vec.
- Trains the linear layer while freezing wav2vec2 on speech-text pair data.
  - e.g., CTC loss

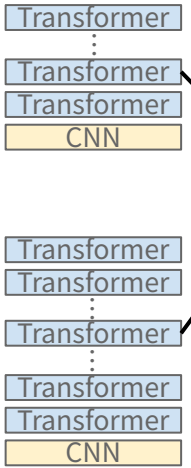




Surprisingly low WER using only 10 min. of pair data.

Table 1: WER on the Librispeech dev/test sets when training on the Libri-light low-resource labeled data setups of 10 min, 1 hour, 10 hours and the clean 100h subset of Librispeech. Models use either the audio of Librispeech (LS-960) or the larger LibriVox (LV-60k) as unlabeled data. We consider two model sizes: BASE (95m parameters) and LARGE (317m parameters). Prior work used 860 unlabeled hours (LS-860) but the total with labeled data is 960 hours and comparable to our setup.

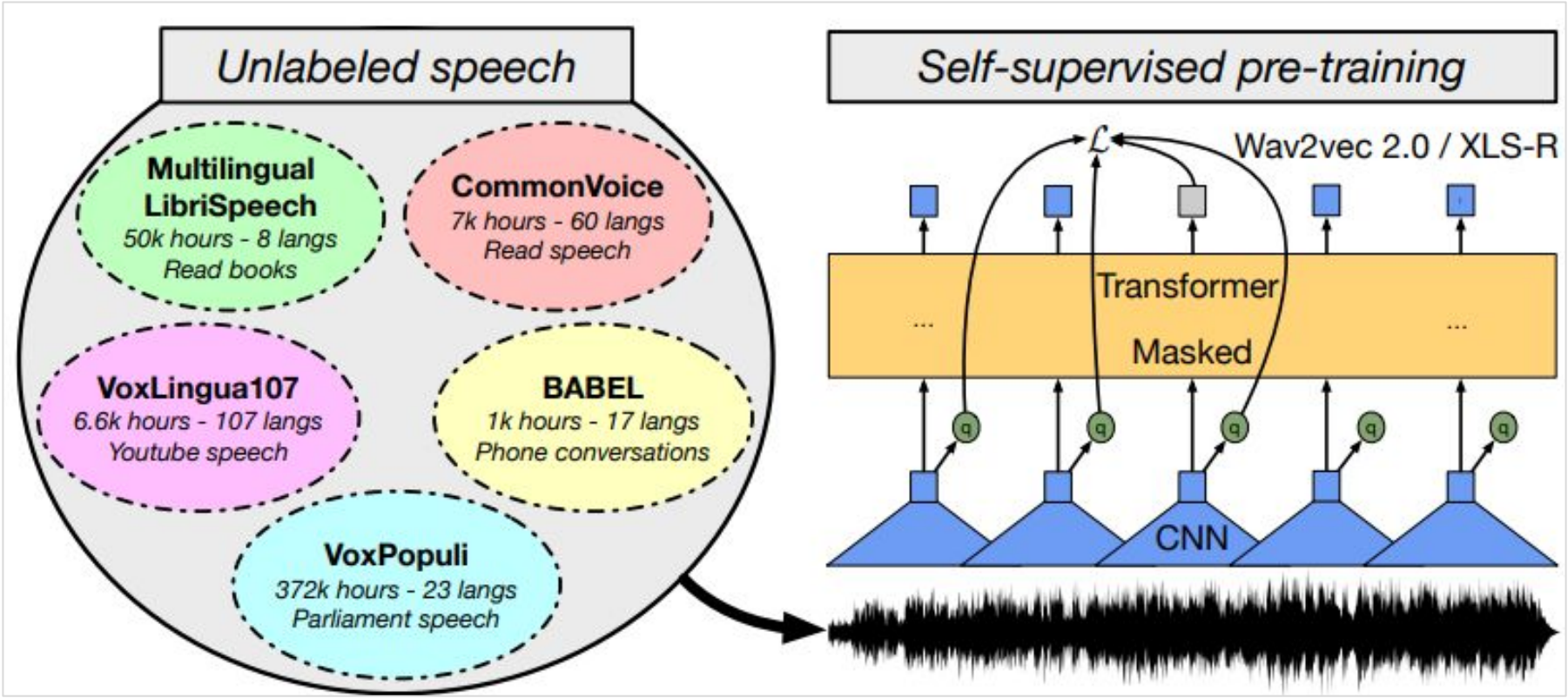
Model	Unlabeled data	LM	dev		test	
			clean	other	clean	other
10 min labeled						
Discrete BERT [4]	LS-960	4-gram	15.7	24.1	16.3	25.2
BASE	LS-960	4-gram	8.9	15.7	9.1	15.6
		Transf.	6.6	13.2	6.9	12.9
LARGE	LS-960	Transf.	6.6	10.6	6.8	10.8
	LV-60k	Transf.	4.6	7.9	4.8	8.2



External language model

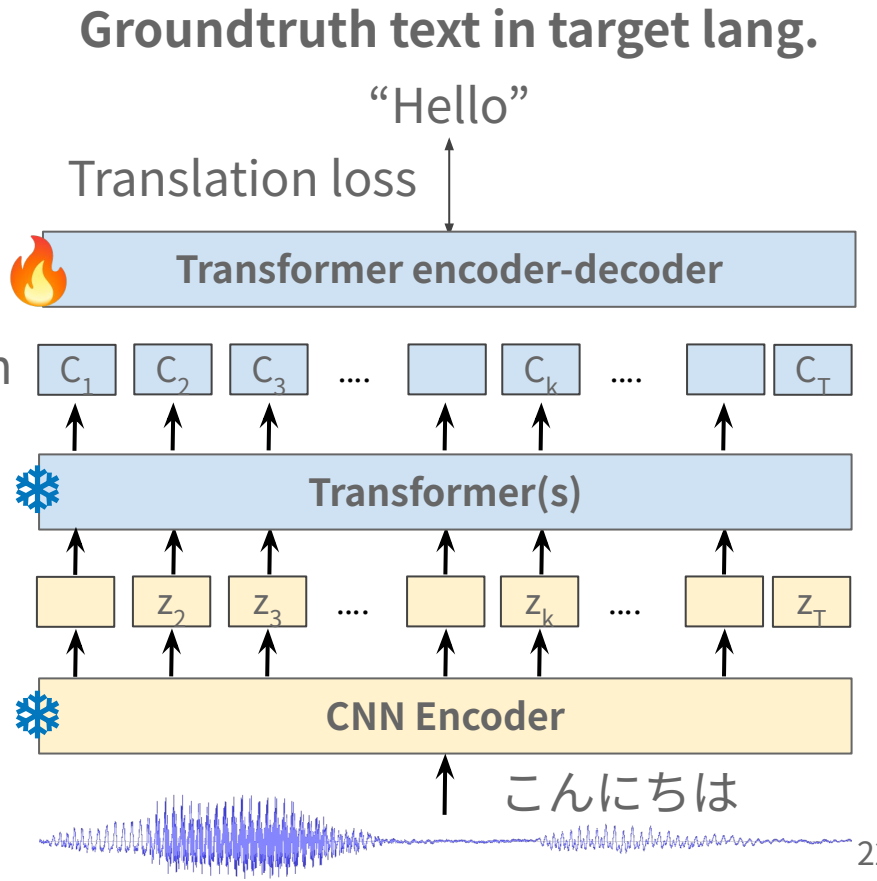
WER

Extending to multi-linguality: XLS-R covering 128 languages



XLS-R → speech-to-text translation

- Pretrains XLS-R on multi-lingual speech-only data.
  - The pretrained model is expected to support multilinguality.
- Adds Transformer encoder-decoder on top of XLS-R.
  - Encoder includes self-attn.
  - Decoder includes cross-attn.
- Trains the Transformer enc-dec.
  - e.g., cross-entropy.



X-Speech to English-text translation

Table 3: Speech translation: results for X → English directions on CoVoST-2 in terms of average BLEU for 21 directions grouped into high/mid/low-resource labeled data directions. Appendix A lists detailed results for all languages.

	high	mid	low	Avg.
Prior work				
XLSR-53 (Conneau et al., 2021)	30.3	11.1	3.2	10.3
VP-100K (Wang et al., 2021b)	27.7	13.2	4.6	11.1
XMEF-En (Li et al., 2021b)	32.4	16.8	4.0	12.4
XMEF-X (Li et al., 2021b)	34.2	20.2	5.9	14.7
This work				
XLS-R (0.3B)	30.6	18.9	5.1	13.2
XLS-R (1B)	34.3	25.5	11.7	19.3
XLS-R (2B)	**36.1**	**27.7**	**15.1**	**22.1**

Multilingual training significantly contributes to multilingual training.

HuBERT (2021)

Overview

* HuBERT feats. are used in the middle of training steps.

- Quantize speech feats.
- Randomly mask CNN outputs.
- Predict the quantization class at masking frames.

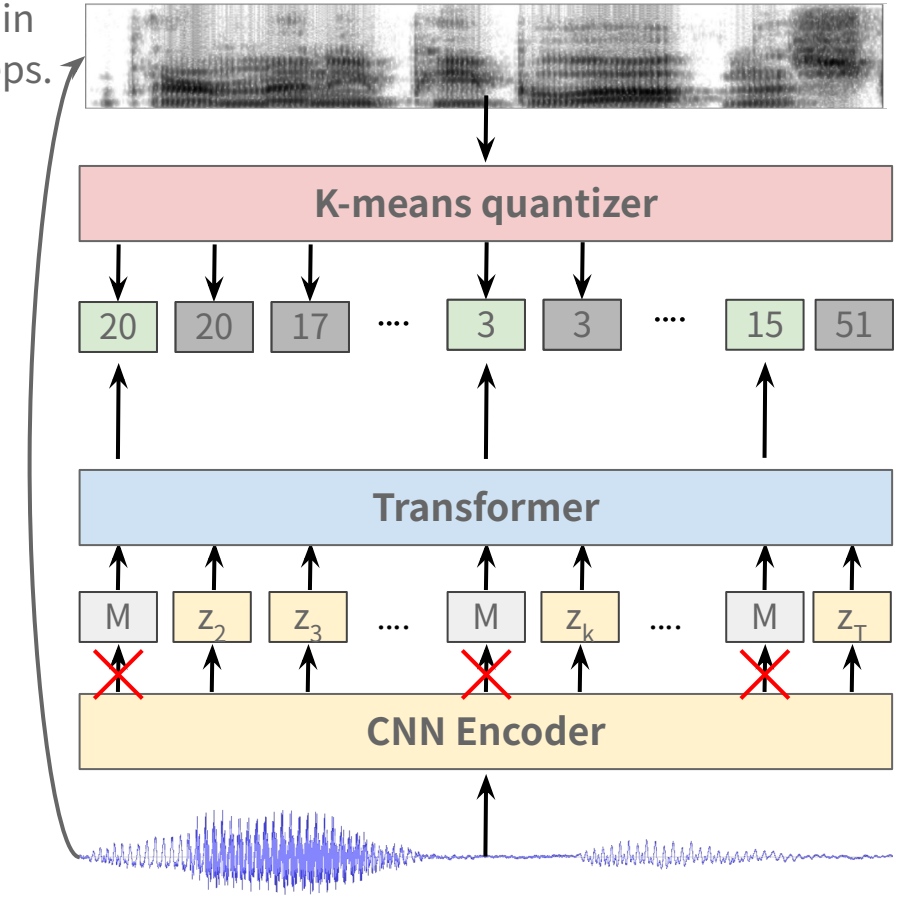
  → Learn to represent "semantic" information from speech.

Training objective

Softmax cross entropy

$$\mathcal{L}_m = \sum_{t \in M} -\log p(y_t \mid X)$$

For all masked frames



Further improvements than wav2vec 2.0

Model	Unlabeled Data	LM	dev-clean	dev-other	test-clean	test-other
		10-min labeled				
DiscreteBERT [51]	LS-960	4-gram	15.7	24.1	16.3	25.2
wav2vec 2.0 BASE [6]	LS-960	4-gram	8.9	15.7	9.1	15.6
wav2vec 2.0 LARGE [6]	LL-60k	4-gram	6.3	9.8	6.6	10.3
wav2vec 2.0 LARGE [6]	LL-60k	Transformer	4.6	7.9	4.8	8.2
HUBERT BASE	LS-960	4-gram	9.1	15.0	9.7	15.3
HUBERT LARGE	LL-60k	4-gram	6.1	9.4	6.6	10.1
HUBERT LARGE	LL-60k	Transformer	4.3	7.0	4.7	7.6
HUBERT X-LARGE	LL-60k	Transformer	4.4	6.1	4.6	6.8

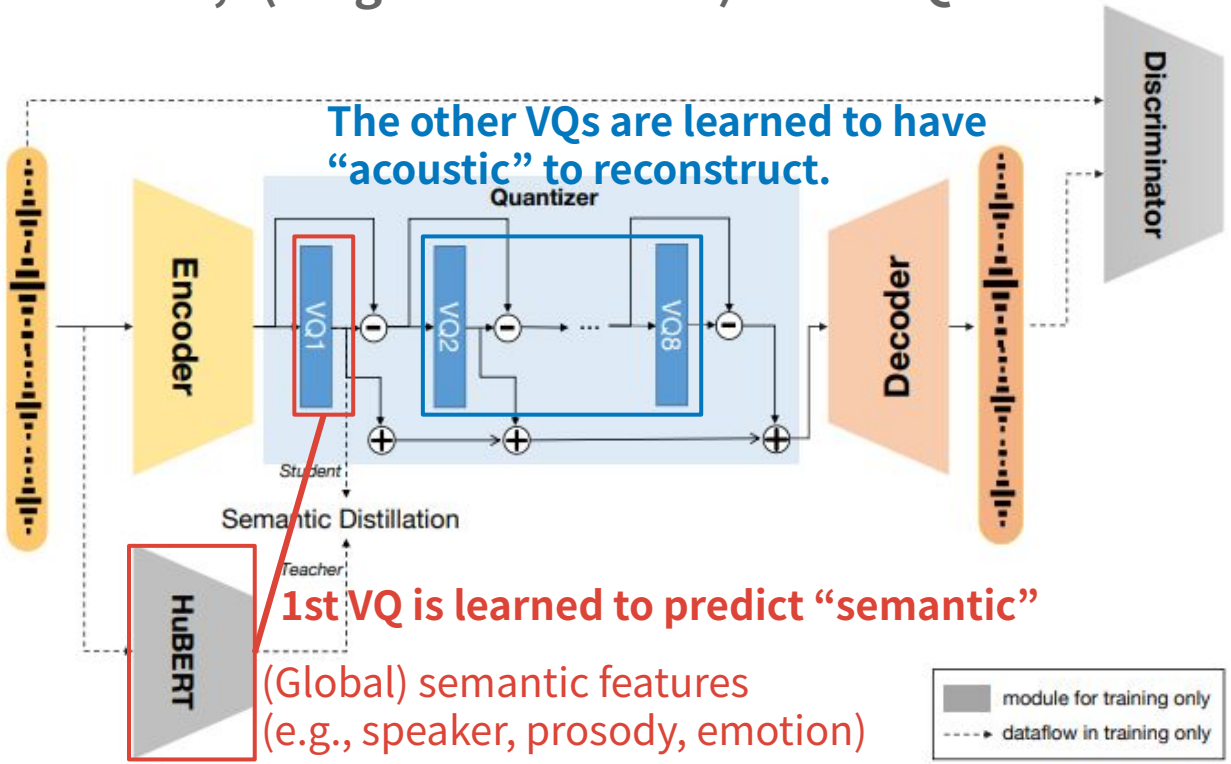
Model size

External language model

WER

Guiding neural speech codec: disentangling “semantic” and “acoustic” (SpeechTokenizer)

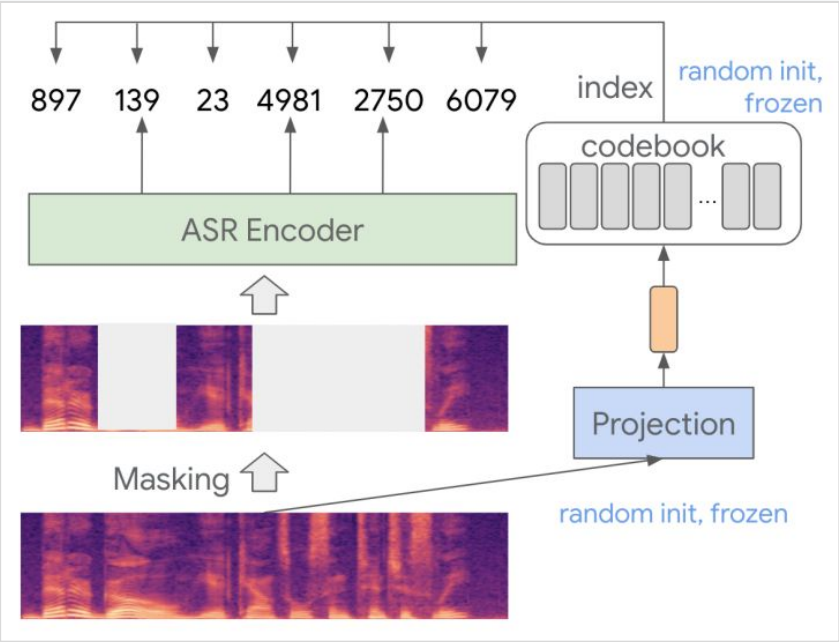
1st VQ for “semantic,” (long-term features) other VQs for “acoustic” (short)



BEST-RQ (from Google)

Random projection to higher dim.

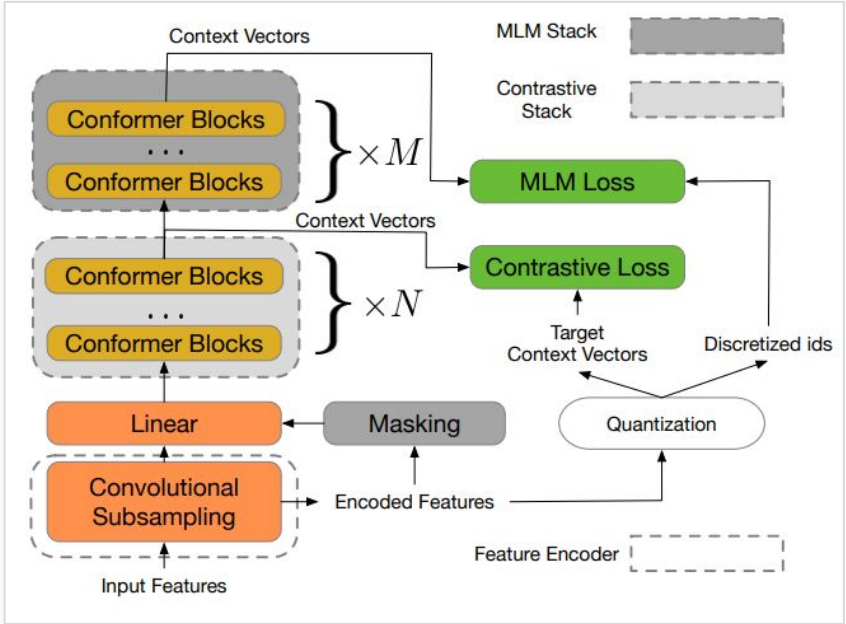
The projection is good at teacher symbols.



https://arxiv.org/pdf/2202.01855

BERT-like training

Masked language modeling learns context.



https://arxiv.org/pdf/2108.06209

(Revisit) Three types of self-supervised learning

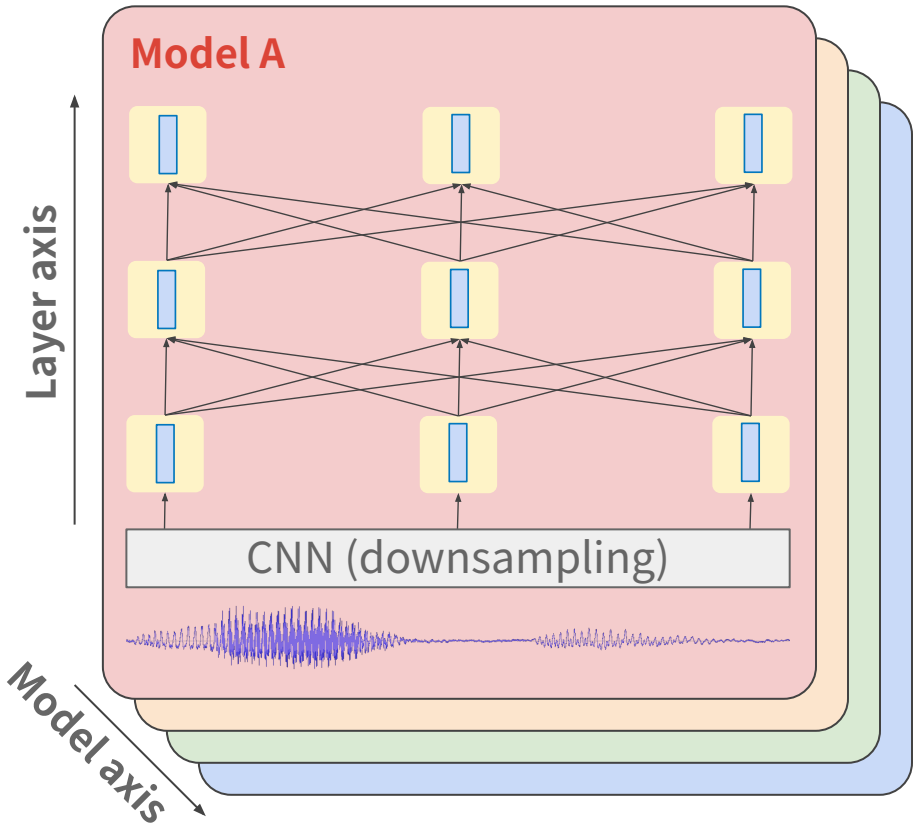
- **Contrastive approaches / 対照型** ➡ Learns discriminative features.
  - Create positive and negative samples.
  - Move the positive sample close and the negative sample far.

- **Predictive approaches / 予測型** ➡ Learns semantic features. (e.g., semantic tokens)
  - (Typically) drop features.
  - Predict the dropped features from undropped ones.

- **Generative approaches / 生成型** ➡ Learns acoustic features. (e.g., acoustic tokens)
  - Adopt generative models.
  - (Re)generate features via the generative models.

What do SSL models learn?

Concepts



Several options for feat. extraction

- Which layer you use
- What model you use

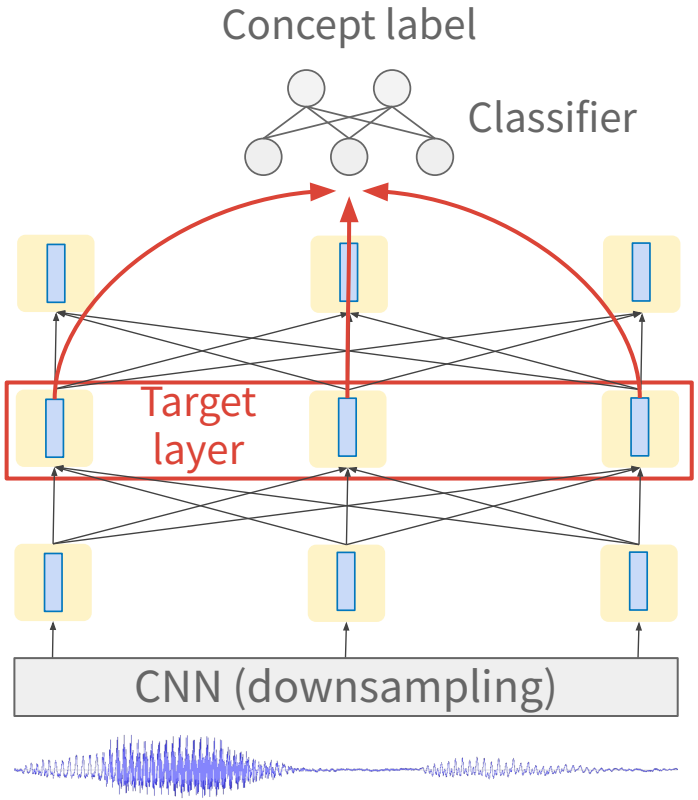
What information does each {layer, model} have?

- phoneme? word?
- speaker? emotion?
- mental state?

Methodologies

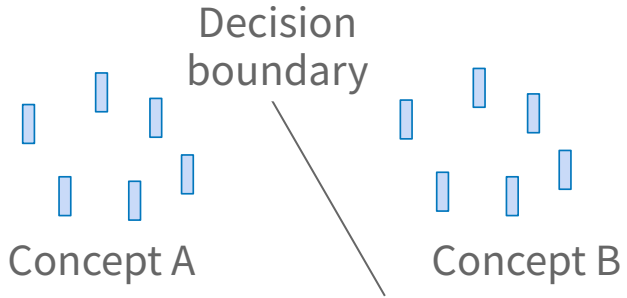
- **Probing**
  - Investigating what specific information is encoded.
- **Similarity**
  - Quantifying how closely related representations are across speech.
- **Factorization**
  - Decomposing representations into interpretable features.
- **(Geometry)**
  - (Characterizing the structure of the representation space.)

Probing



Trains simple (linear) classifier

- Input: features of target {layer, model}
- Output: concept label

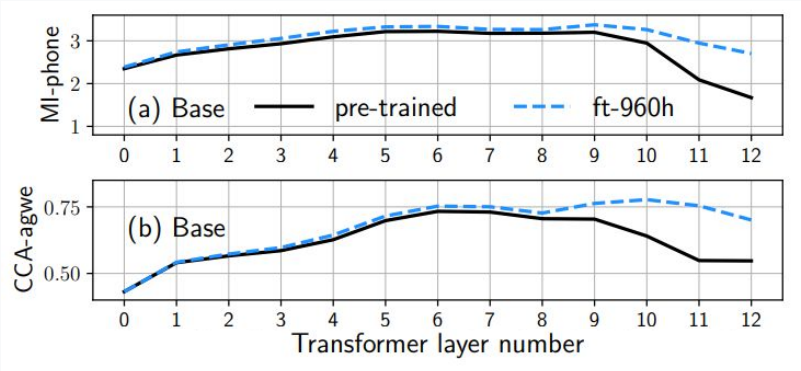


High classification accuracy -> The target {layer, model} can discriminate the target concepts.

Mutual information from representations to phoneme (or word) labels on wav2vec2.

Mutual information to phonemes

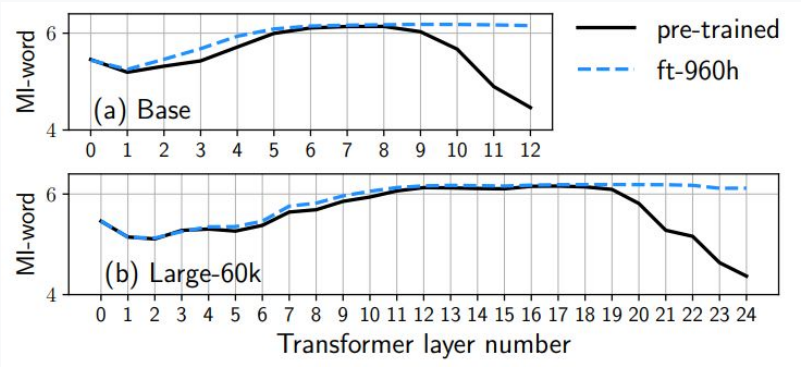
- Weak information in all layers
- ASR finetuning (ft-960h) enhances in deeper layers



https://arxiv.org/pdf/2107.04734

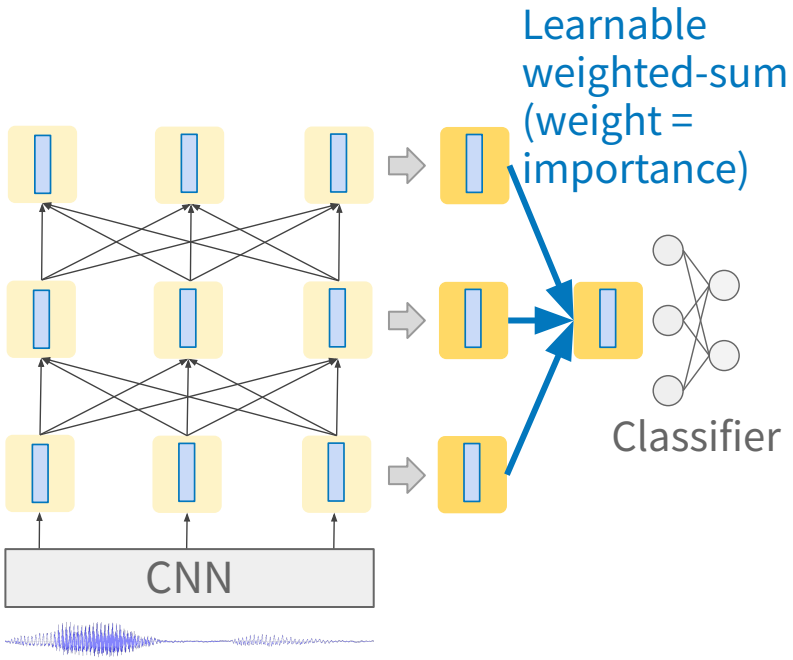
Mutual information to words

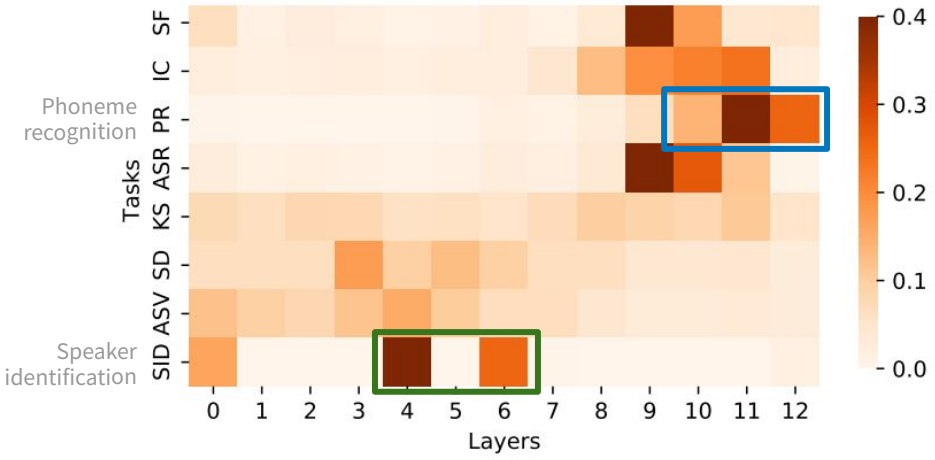
- Strong information in mid. layers
- Difference by model size
- ASR finetuning enhances in deeper laeyers.



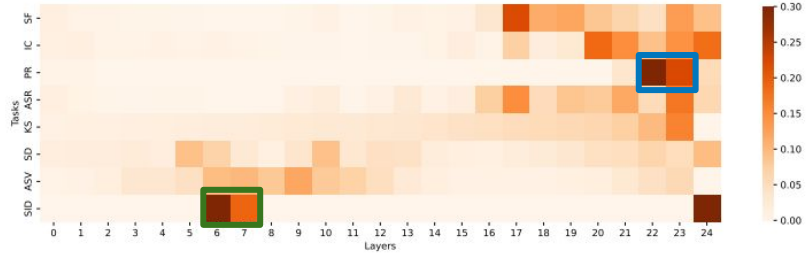
https://arxiv.org/pdf/2609.10434

Weighted sum





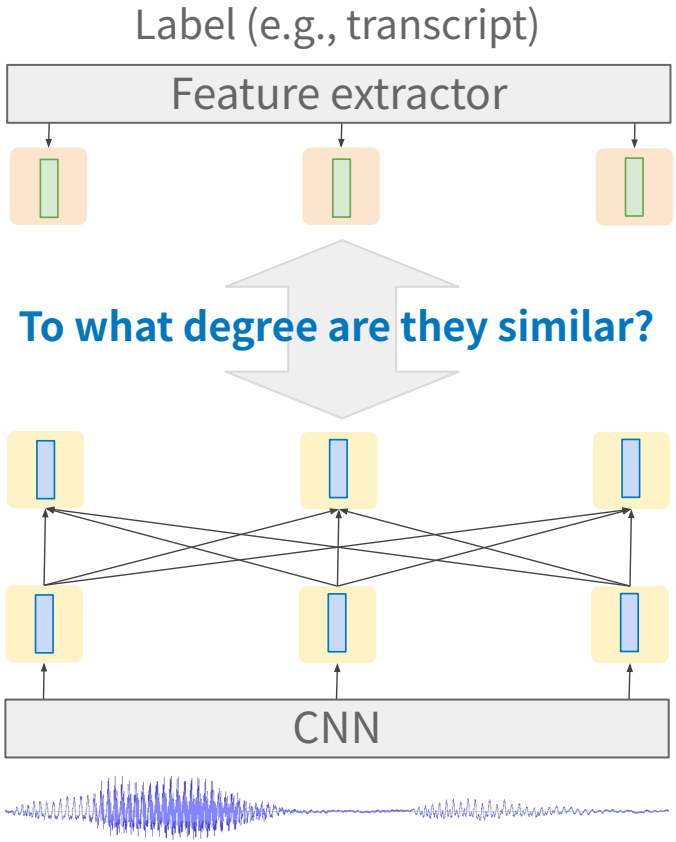
(b) WavLM Base+



(d) WavLM Large

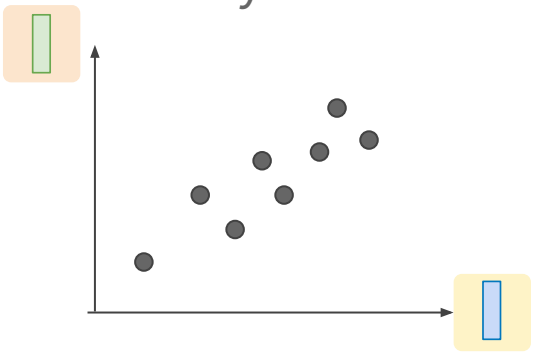
https://arxiv.org/abs/2110.13900

Similarity



Calculates similarity between two features

- Input: features of target {layer, model}
- Output: similarity score



High score -> The target {layer, model} correlates with the label-derived features, e.g., word embedding.

canonical-correlation analysis (CCA)

Correlation analysis / 相関分析

Extent to which a pair of **scalar values** are linearly related.

Covariance of (x, y)

$$\rho = \frac{\mathrm{Cov}(x, y)}{\sigma_x \sigma_y}$$

Stdev. of scalar x Stdev. of scalar y

CCA / 正準相関分析

Correlation for **vector values**. Vector values are dimensionally reduced into scalar.

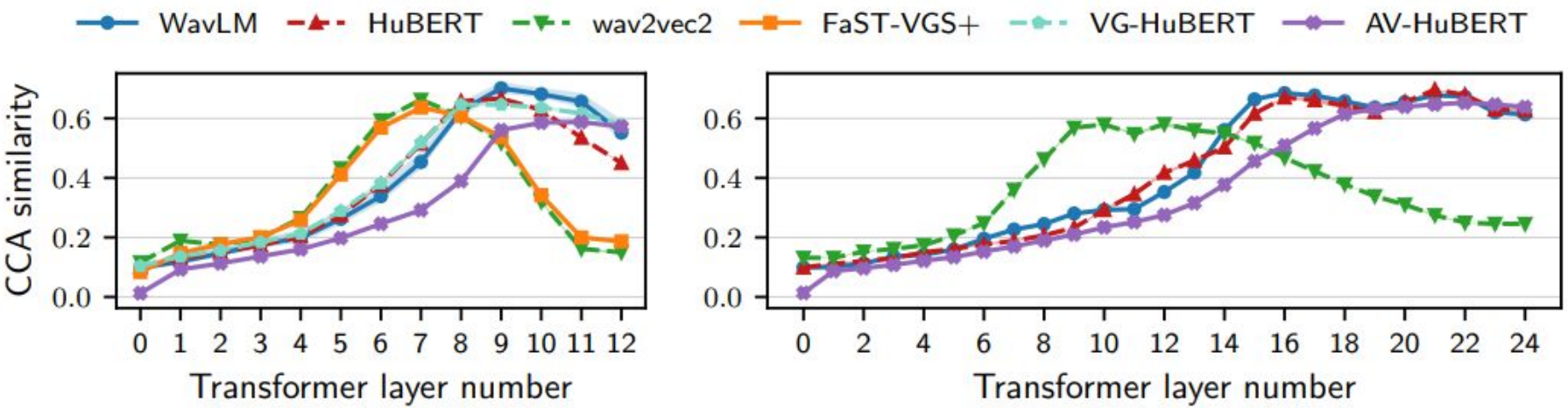
(Dim. reduction vectors a, b are learned to maximize rho.)

Covariance of dim.-reduced (**x**, **y**)

$$\rho = \frac{\mathrm{Cov}(\boldsymbol{a}^\top \boldsymbol{x}, \boldsymbol{b}^\top \boldsymbol{y})}{\sqrt{\mathrm{Var}(\boldsymbol{a}^\top \boldsymbol{x})\mathrm{Var}(\boldsymbol{b}^\top \boldsymbol{y})}}$$

Stdev. of dim.-reduced vector **x** Stdev. of dim.-reduced vector **y**

Embeddings vs. word vectors



(a) *CCA similarity between mean-pooled S3M word segment representations and word ID one-hot vectors.*

- SSL models typically have word-correlated representations in mid. layers.
  - VG-HuBERT (base): 6, WavLM (base): 9
- Even if the same model series, e.g., WavLM (base, large), the peak layers differ.

https://arxiv.org/pdf/2307.00162

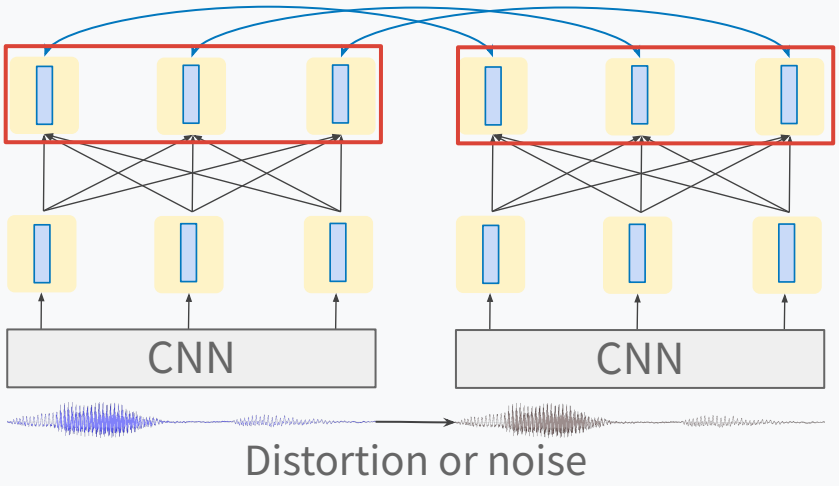
Similarity

Contrastive similarity

Similarity between contrastive pairs,
e.g., robustness to noise and distortion.

"FROM PRETRAINING TO ROBUSTNESS: BENCHMARKING SSL MODELS FOR NOISE-ROBUST SPEECH EMOTION RECOGNITION", 2026

To what degree are they similar?

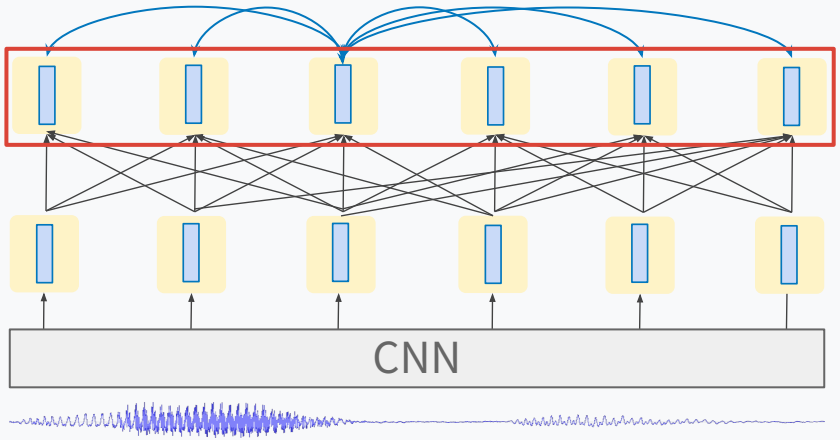


Self similarity

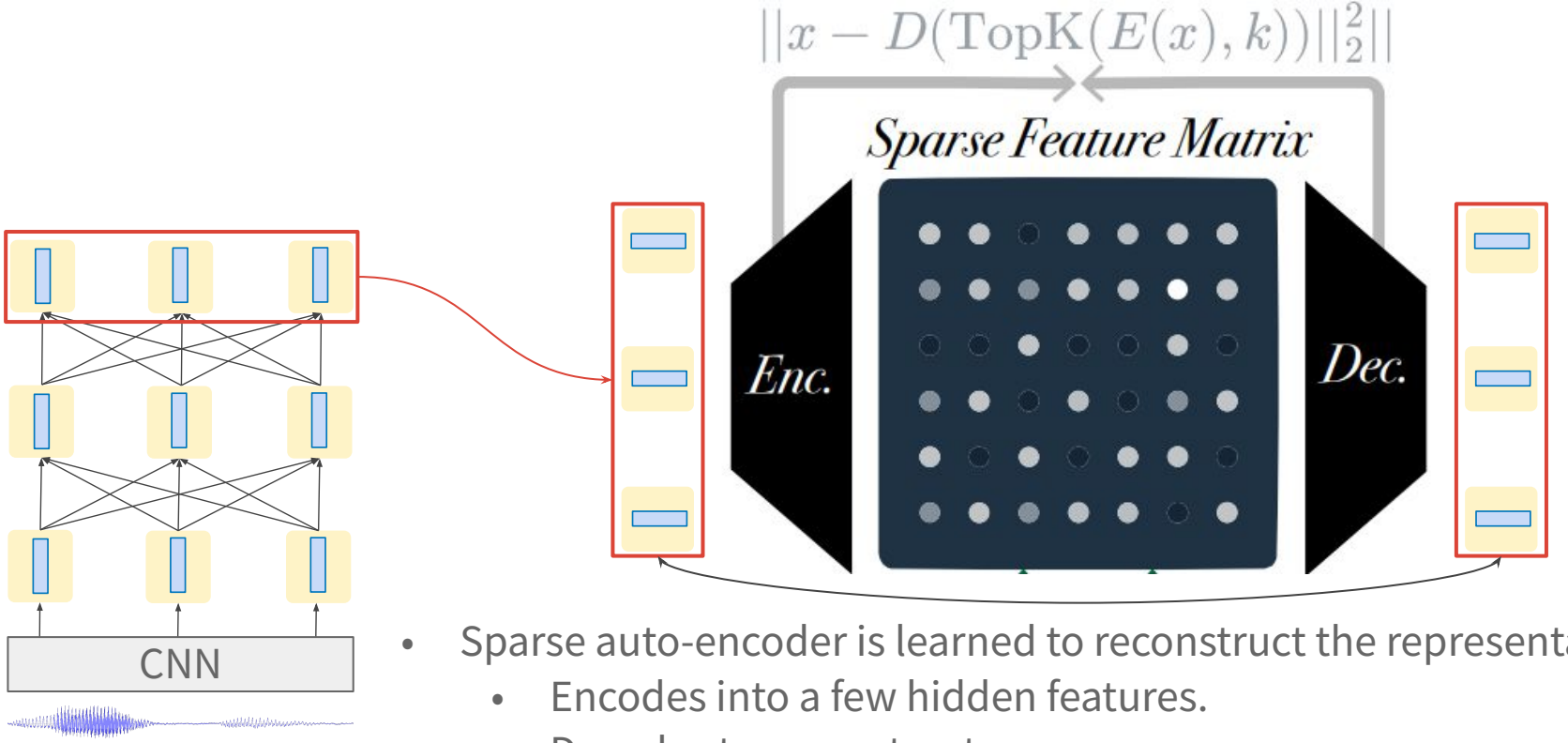
Similarity among frames in one sequence,
e.g., distinguishability on syllables.

"Sylber 2.0: A Universal Syllable Embedding", 2026

To what degree are they similar?



Factorization: sparse auto-encoder



- Sparse auto-encoder is learned to reconstruct the representations.
  - Encodes into a few hidden features.
  - Decodes to reconstruct.
- **The hidden features become factors of the representations.**

Conclusion

Conclusion

- Self-supervised learning (SSL)

- Basic architecture
  - wav2vec2, XLS-R, HuBERT, BEST-RQ
  - fine-tuning, neural audio codec

- Analysis of internal representations
  - Probing, similarity, factorization